Expansion Is Not Closure: Settlement Stack Competition and the Conditional Relevance of External Connective Assets

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Abstract

This paper examines whether the expansion of institutionally connected digital infrastructures implies progressive settlement closure, or whether economically meaningful residual overflow may remain across competing settlement stacks. It argues that expansion and closure are analytically distinct. Stablecoins, tokenized deposits, CBDCs, internal treasury rails, omnibus structures, and orchestration mechanisms may enlarge the settlement field, but closure is the stronger condition in which residual overflow has been compressed to economic immateriality.

The paper develops a stack-competition framework for analyzing this distinction. It classifies the main structural sources of residual overflow across jurisdictional, ledger, balance-sheet, collateral, timing, institutional, compliance, and stress-related dimensions, and compares the major closure mechanisms according to the boundaries they internalize and the unresolved edges they leave behind. It further argues that residual overflow alone does not establish the relevance of any external connective asset. A stronger claim requires that residual overflow remain economically material, generate held-form demand in inventory, collateral, liquidity-buffer, or waiting-liquidity form, and survive institutional selection filters such as governance, compliance, liquidity-depth, distribution, operational, and incentive constraints.

The contribution of the paper is therefore not to argue that external connective assets necessarily persist, nor that expanding infrastructure guarantees their obsolescence. It is to specify the conditions under which their relevance remains structurally possible, and the conditions under which that relevance weakens or is invalidated by effective closure, full-coverage substitution, efficiency compression, selection failure, or absence of retained-demand evidence. Although an XRP-adjacent context appears as an illustrative stress extension, the paper's contribution is methodological rather than asset-specific. It offers a deliberately restrictive framework for evaluating external connective assets under closure competition in institutionally embedded digital finance.

1. Introduction

Digital financial infrastructure is expanding rapidly across multiple institutional layers. Stablecoins, tokenized deposits, shared ledgers, programmable collateral systems, treasury platforms, and emerging CBDC architectures are broadening the range of environments in which digitally connected assets may potentially operate. But expansion enlarges the settlement field; closure is the stronger claim that economically meaningful residual overflow has been compressed to immateriality. The central question

of this paper is therefore not whether infrastructure is expanding, but whether competing settlement stacks are actually closing the relevant boundary-crossing problem.

Paper 1 argued that institutionally connected digital assets should not be evaluated through visible usage alone, but through the stronger question of whether institutional environments create reasons to retain them in inventory, collateral, liquidity-buffer, or waiting-liquidity form. It also separated domain expansion, asset selection, and asset indispensability as analytically distinct layers. The present paper extends that framework by examining one of the main intervening questions between expansion and indispensability: whether the growth of competing institutional settlement stacks absorbs the frictions that would otherwise sustain demand for an external connective asset.

This distinction matters because infrastructure growth and settlement closure are not the same phenomenon. Expansion may enlarge the available settlement surface by increasing the number of recognized rails, instruments, and institutional workflows. Closure, by contrast, requires that economically meaningful frictions be internalized to the point that external bridge-like intermediation becomes unnecessary. A system may therefore become more digitally connected, more compliant, and more operationally sophisticated while still leaving unresolved boundaries across jurisdictions, ledgers, balance sheets, collateral regimes, timing structures, and institutional perimeters. Throughout this paper, residual overflow refers not to any remaining imperfection as such, but only to frictions that remain economically material after competing closure mechanisms have operated.

The argument of this paper is not that expanding infrastructure automatically strengthens the case for an external connective asset, nor that it automatically eliminates it. Both simplifications are inadequate. What matters is whether competing closure mechanisms can compress residual overflow to the point that an external connective asset becomes economically immaterial. If they can, the retained-demand case weakens sharply. If they cannot, then the relevance of an external connective asset may persist, not because infrastructure failed to develop, but because development did not culminate in full settlement closure.

This framing also clarifies the relation between overflow and retained demand. If residual overflow survives, it may generate held-form demand through several channels already identified in Paper 1. Boundary-crossing workflows may require positioned balances and thus generate inventory demand. Incomplete collateral portability may generate collateral demand. Execution uncertainty, mismatch, or stress may generate liquidity-buffer demand. Temporal gaps in conditional or staged processes may generate waiting liquidity. Paper 2 therefore does not replace the retained-demand framework. It specifies one of the principal structural environments under which retained demand may or may not emerge.

The paper proceeds from the observation that modern settlement infrastructures compete not as a single convergent system, but as partially overlapping stacks. Stablecoins, tokenized deposits, CBDCs, internal treasury rails, omnibus structures, prime arrangements, and orchestration layers each attempt to internalize specific forms of settlement friction. But their closure capacities are not identical. Some are strong within a defined institutional perimeter while weak across perimeters. Some improve transferability while remaining constrained by liability boundaries, eligibility rules, or legal jurisdictions. Some compress timing frictions without eliminating balance-sheet separation. For that reason, expansion should not be confused with full coverage, and interoperability should not be assumed to imply closure.

The contribution of this paper is fivefold. First, it introduces a stack-competition framework for evaluating whether residual bridge-like overflow remains after competing closure mechanisms operate. Second, it classifies the main structural sources of residual overflow across jurisdictional, ledger, balance-sheet, collateral, timing, institutional, compliance, and stress-related dimensions. Third, it compares the major closure mechanisms not as abstract innovations, but as stack-specific attempts to internalize particular forms of friction, each with different strengths and different unresolved boundaries. Fourth, it shows that overflow and selection must be analytically separated, because governance, compliance, liquidity-depth, distribution, operational, and incentive constraints may block realized relevance even where residual space persists. Fifth, it specifies the conditions under which an external connective asset may still matter, as well as the conditions under which that relevance is weakened or invalidated by full-coverage alternatives, internalized routing, or efficiency compression.

Although XRP remains the primary motivating case in the background, the contribution of this paper is not XRP-specific advocacy. Nor is it a price thesis. Its purpose is to provide a conditional analytical framework for evaluating whether external connective assets retain a meaningful role once digital financial infrastructures become more institutionally embedded, more internally optimized, and more densely contested by alternative closure mechanisms.

This paper therefore defines the second layer of the retained-demand audit series. Paper 1 defined retained demand by type; the present paper asks whether competing settlement stacks close the residual boundary conditions that could otherwise sustain external connective-asset relevance. Later papers extend this sequence to required liquidity density, stack-level compression and bypass, evidence-gated sizing, and backend retained-demand location after user abstraction.

The remainder of the paper proceeds as follows. Section 2 explains why infrastructure expansion should not be treated as equivalent to settlement closure. Section 3 classifies the main structural sources of residual bridge-like overflow. Section 4 examines competing closure mechanisms, including stablecoins, tokenized deposits, CBDCs, internal treasury rails, omnibus structures, and orchestration mechanisms. Section 5 analyzes layer-lock and selection constraints. Section 6 specifies the conditions under which external connective assets may still matter, together with their invalidation conditions. Section 7 briefly extends the framework to an XRP-adjacent stress context. Section 8 concludes and situates the argument as a bridge to a later density- and sizing-oriented inquiry.

2. Why Expansion Does Not Necessarily Mean Closure

The expansion of digital financial infrastructure is often treated as if it were equivalent to progressive settlement closure. As more institutions adopt stablecoins, tokenized deposits, treasury platforms, shared ledgers, and programmable settlement tools, it is tempting to infer that the space for an external connective asset must necessarily shrink. Yet this inference is too fast. Expansion and closure describe different phenomena and must be separated analytically.

Expansion refers to the widening of the institutional settlement surface. More rails become available. More instruments become recognized. More workflows become digitally connected. More participants gain access to technically or legally sanctioned forms of transfer, coordination, and settlement. These developments are significant, but they do not by themselves establish that the frictions arising across those expanding domains have been fully absorbed.

Closure is a stronger condition. A settlement environment approaches closure when the relevant frictions across domains are internalized to the point that external bridge-like intermediation becomes economically unnecessary. This requires more than digitalization or scale. It requires that cross-jurisdictional movement, cross-ledger transfer, balance-sheet separation, collateral incompatibility, temporal mismatch, and institutional boundary effects be compressed sufficiently that residual overflow no longer carries material economic weight.

This distinction is critical because financial infrastructures do not expand as a single unified system. They expand as partially overlapping stacks with different legal structures, governance rules, liability forms, liquidity profiles, access conditions, and operational incentives. A stablecoin network may scale globally while remaining dependent on issuer-specific structures and jurisdictional constraints. A tokenized deposit system may be highly efficient within a banking perimeter while remaining limited across banking perimeters. A CBDC may offer strong domestic settlement finality while remaining weak in cross-border portability. An orchestration layer may optimize routing without dissolving the underlying liability boundaries of the systems it connects. In each case, expansion may be substantial while closure remains incomplete.

For this reason, the relevant analytical question is not whether infrastructure is improving, but what kind of frictions that improvement actually absorbs. Some frictions may indeed shrink materially. Others may be displaced, transformed, or re-expressed at different layers. Increased institutional sophistication may compress visible inefficiencies while leaving deeper forms of fragmentation intact. In some cases, digitalization may even intensify segmentation by hardening governance perimeters, eligibility rules, or system-internal incentives.

The possibility of incomplete closure also explains why expanding infrastructure should not be read in either of two overly simple ways. The first simplification is that more infrastructure necessarily eliminates the need for an external connective asset. The second is that more infrastructure necessarily proves the future importance of such an asset. Both views are analytically weak because both infer too much from expansion alone. The real issue is whether competing closure mechanisms are sufficiently comprehensive to absorb economically meaningful overflow after they operate.

Accordingly, this paper treats expansion as evidence of a larger competitive settlement field, not as decisive evidence of closure. The growth of stablecoins, tokenized deposits, CBDCs, internal treasury rails, omnibus arrangements, and orchestration mechanisms matters because each represents an attempt to close particular forms of friction from within a given stack. But the success of those attempts cannot be assumed in the abstract. Their effectiveness must be examined in relation to the particular boundaries they are able to internalize and the residual overflow they leave behind.

Once this distinction is established, the next analytical step becomes clearer. If expansion does not itself prove closure, then one must identify where residual bridge-like overflow continues to arise. The relevant task is therefore not to celebrate infrastructure growth or to defend a bridge thesis by default, but to classify the structural sources of overflow that may survive competing closure mechanisms. That is the purpose of the next section.

3. Residual Sources of Bridge-Like Overflow

If expansion does not itself imply closure, the next analytical question is where residual bridge-like overflow continues to arise. The answer is not a single technical gap, but a set of recurrent structural boundaries that may persist even as digital financial infrastructures become more advanced, more institutionalized, and more internally optimized. Residual overflow does not denote any remaining imperfection as such. It refers only to frictions that remain economically material after available internal closure mechanisms operate. In this paper, "economically material" does not mean merely formally unresolved. It means that the remaining boundary continues to impose non-trivial cost, delay, capital lock-up, collateral impairment, or execution vulnerability under actual institutional workflows. The relevant issue, therefore, is not whether any boundary still exists in formal terms, but whether that boundary still carries enough economic weight to affect settlement behavior, balance-sheet positioning, or operational design.

The first source is jurisdictional overflow. Digital infrastructures do not operate in a legally neutral space. Even when transfer technologies improve, settlement claims continue to move across non-identical legal, regulatory, monetary, and supervisory environments. Finality standards differ. Eligible instruments differ. Capital and compliance constraints differ. Currency domains remain distinct. For this reason, efficient rails within one jurisdiction do not automatically imply seamless closure across jurisdictions. Where value must still traverse non-identical legal and monetary perimeters, residual overflow may remain economically material. This overflow becomes economically material where claims must cross non-identical legal and monetary perimeters at meaningful scale without reliable reciprocal closure, offsetting, or end-to-end institutional absorption.

The second source is ledger or platform overflow. Digitally connected finance is not converging onto a single unified ledger. Value remains issued, managed, and transferred across multiple technological and institutional environments with distinct settlement rules, governance logics, identity requirements, and asset representations. Even where interoperability improves, cross-platform movement may still require staged transfer, duplicate positioning, or additional coordination rather than constituting a frictionless single-state process. It becomes materially relevant where cross-platform movement cannot be reduced to internal book-entry substitution and instead requires staged transfer, duplicate positioning, or additional settlement coordination.

The third source is balance-sheet overflow. Many financial instruments remain inseparable from the liability structures and credit boundaries of their issuers. Tokenized deposits, for example, may be digitally transferable in form while still remaining economically bounded by the issuing bank's balance sheet. Similar limits may arise in omnibus structures, internal treasury environments, or institution-specific settlement systems. What appears digitally connected at the interface level may therefore remain segmented at the liability level. Where value transfer requires movement across non-identical balance-sheet perimeters, residual overflow may persist even when the visible infrastructure appears highly integrated. It becomes economically material where liability boundaries remain binding under actual funding, settlement, or credit conditions rather than being neutralized through integrated balance-sheet arrangements.

The fourth source is collateral mobility overflow. Collateral does not become universally portable merely because it is tokenized or digitally represented. Eligibility criteria, rehypothecation rules, custody

arrangements, haircut practices, legal enforceability, and risk-management standards can differ materially across infrastructures and counterparties. An asset that functions as recognized collateral in one environment may not move frictionlessly into another, or may do so only with delay, operational constraint, or value impairment. Where collateral portability remains incomplete, there may be continuing demand for intermediate forms of value positioning, reserve maintenance, or connective transfer. It becomes materially significant where portability limits force reserve maintenance, haircut-bearing transfer, delayed reuse, or constrained eligibility across infrastructures.

The fifth source is timing and synchronization overflow. Even in digitally accelerated systems, time remains inside the settlement process. Workflows still involve sequencing, pending conditions, asynchronous ledgers, market-hour differences, reconciliation cycles, conditional release logic, and delayed confirmation structures. As long as processes do not resolve in a fully synchronized manner, capital may remain staged, trapped, prefunded, or buffered across time. This temporal dimension is analytically important because it creates waiting-liquidity conditions even where underlying infrastructures are otherwise efficient. It becomes economically material where asynchronous processes require prefunding, trap capital in staged form, or create non-trivial balance-sheet exposure during incomplete settlement sequences.

The sixth source is institutional boundary overflow. Financial infrastructures are embedded in organizational perimeters defined by governance, internal risk controls, approval flows, custody models, and operating mandates. A system may close frictions effectively within one firm, bank, market operator, or consortium while leaving significant frictions at the edge of that perimeter. Internal efficiency is therefore not equivalent to inter-institutional closure. Residual overflow may remain precisely at the point where different institutional domains meet, even if each domain is internally optimized. It becomes materially relevant where internal optimization within one perimeter does not remove the cost, delay, or uncertainty of coordination across institutional boundaries.

A seventh source, closely related to the others, is compliance or eligibility overflow. Access to particular rails, assets, or transfer channels is often asymmetric. Participation may depend on regulatory status, counterparty type, jurisdiction, custody arrangement, documentation standards, or operational readiness. These asymmetries can restrict direct reuse or transfer even where the underlying technology permits it. In such cases, the limiting factor is not merely technical incompatibility but differentiated admissibility. This can preserve structurally important discontinuities between nominally connected infrastructures. It becomes economically material where differentiated admissibility restricts direct transfer, reuse, or recognition across otherwise connected infrastructures.

An eighth source is stress or execution overflow. In normal conditions, orchestration, netting, and internal routing may compress large parts of the settlement problem. But stress conditions can reintroduce frictions that are muted in ordinary operation. Slippage, failed execution, widened spreads, conditional liquidity withdrawal, operational delay, or sudden increases in margin and reserve needs may create temporary but economically meaningful overflow. This matters because a closure mechanism should not be judged only by its performance under calm conditions if the system's real demands emerge under discontinuity. It becomes materially relevant where stress conditions reactivate liquidity, margin, execution, or settlement frictions that remain muted in normal-state operation.

These categories should not be treated as mutually exclusive. In practice, a single workflow may involve several layers of overflow at once: a cross-jurisdictional transfer may also cross ledgers, balance sheets, collateral regimes, timing sequences, and institutional boundaries. The analytical purpose of classification is therefore not to isolate perfectly pure cases, but to identify the main structural channels through which residual bridge-like frictions may survive infrastructure expansion.

This classification also clarifies the relation between overflow and retained demand. Different forms of overflow are likely to activate different held-form requirements. Jurisdictional, ledger, and institutional boundary overflow may generate inventory demand through the need for positioned balances. Collateral mobility overflow may generate collateral demand. Timing and synchronization overflow may generate waiting liquidity. Stress and execution overflow may generate liquidity-buffer demand. The significance of residual overflow, therefore, does not lie simply in the fact that some friction remains. It lies in the possibility that specific forms of friction continue to produce economically meaningful reasons to hold value in advance.

At the same time, the existence of residual overflow does not establish that any specific external connective asset will be selected. Overflow identifies a structural space, not a winner. Whether an external asset remains relevant depends on the comparative effectiveness of competing closure mechanisms, the admissibility of the asset within institutional workflows, the depth and reliability of its liquidity, the simplicity of its operational integration, and the incentives of the systems involved. The next step, therefore, is not to infer selection from overflow, but to examine the closure mechanisms that compete to absorb it.

4. Competing Closure Mechanisms

Once the main sources of residual overflow have been identified, the next question is how far competing closure mechanisms can actually absorb them. The relevant issue is not whether stablecoins, tokenized deposits, CBDCs, internal treasury rails, omnibus structures, or orchestration layers are useful. Plainly, they are. The issue is whether their usefulness amounts to sufficiently comprehensive closure across the boundaries that matter. Closure mechanisms should therefore be evaluated not as abstract innovations, but as stack-specific attempts to internalize particular forms of settlement friction. The analytical comparison that follows asks three questions of each mechanism: where it is strongest, across which boundaries it remains weakest, and what primary unresolved boundary it tends to leave behind.

Stablecoins are strongest where the settlement problem is one of transferability across multiple platforms and market environments without depending entirely on traditional correspondent pathways. They can provide a relatively portable instrument across a broad set of digitally connected contexts, and in that sense they offer one of the most visible closure advances against certain forms of ledger or platform overflow. They are weaker, however, across boundaries defined by issuer structure, redemption architecture, jurisdictional admissibility, and regulatory treatment. Their primary unresolved boundary is therefore not simple transferability but the persistence of issuer-bound, compliance-bound, and jurisdiction-bound constraints beneath the surface of portability.

Tokenized deposits are strongest within bank-centered environments where the relevant counterparties already operate inside a compatible institutional and regulatory perimeter. They bring programmability, operational efficiency, and digitally improved transferability to bank liabilities while preserving familiar

balance-sheet structures. They are weaker, however, across bank balance sheets, across non-identical regulatory perimeters, and across institutional contexts in which liability boundaries remain binding. Their primary unresolved boundary is therefore balance-sheet overflow: what appears digitally transferable at the interface level may remain economically segmented at the liability level.

CBDCs are strongest within a sovereign monetary domain where public-money status and official settlement finality provide a uniquely powerful closure capacity. In domestic contexts, this can materially reduce layers of settlement uncertainty and intermediary dependence. They are weaker, however, across jurisdictions, across heterogeneous access designs, and across interfaces with non-sovereign market infrastructures. Their primary unresolved boundary is therefore the boundary between monetary-domain closure and cross-domain portability: strong closure inside a sovereign perimeter does not automatically translate into seamless closure across sovereign perimeters.

Internal treasury rails and institution-specific cash management systems are strongest where the problem is internal coordination within the organizational perimeter of a firm, bank, or financial group. They can reduce prefunding, improve visibility, compress operational inefficiencies, and optimize capital deployment inside the institution. They are weaker, however, once workflows cross into other legal entities, market infrastructures, banking systems, or jurisdictions. Their primary unresolved boundary is therefore inter-institutional overflow: internal optimization does not by itself dissolve friction at the edges between organizational perimeters.

Omnibus structures, prime brokerage layers, and custody-based aggregation models are strongest as interface solutions. They simplify access, reporting, execution, and certain timing frictions by centralizing operational experience within an intermediary layer. They are weaker, however, where the underlying legal, ownership, balance-sheet, or rehypothecation boundaries remain decisive. Their primary unresolved boundary is therefore representational closure without full structural closure: they can make movement look smoother without fully removing the deeper segmentation beneath the structure.

Netting, orchestration, synchronization, and liquidity-saving mechanisms are strongest where the settlement problem can be reduced by minimizing the amount of movement or liquidity that must occur in the first place. They can be extremely powerful, and they represent one of the most serious invalidating pressures on a persistent external connective-asset thesis. They are weaker, however, where obligations are not easily offsettable, processes are not legally or operationally linkable, or participant access remains asymmetric. Their primary unresolved boundary is therefore not the persistence of formal friction as such, but the persistence of non-compressible friction under hard legal, institutional, or balance-sheet boundaries. Messaging coordination, routing intelligence, or credit-line based internalization may conceal certain connective functions, but they do not automatically merge liability structures or neutralize boundary dependence.

These mechanisms do not merely compete with one another; they also interact. A stablecoin may operate inside an omnibus structure. A tokenized deposit may be routed through an orchestration layer. A CBDC interface may connect to bank settlement systems. A treasury rail may reduce internal frictions while relying on an external network for boundary crossing. Because of this, the relevant comparison is not between isolated technologies, but between composite closure capacities. The key analytical question is therefore not which mechanism is most advanced in general, but which boundaries each mechanism can actually internalize, under what institutional conditions, and with what unresolved edges left over.

This is why the expansion of multiple closure mechanisms should not be read as automatic proof of full settlement closure. In one sense, their expansion shows that the market is actively trying to internalize known frictions. In another sense, however, the coexistence of many closure attempts may itself indicate that no single mechanism closes all relevant boundaries equally well. Different stacks may solve different problems while reproducing others in altered form. Closure is therefore better understood as uneven, layered, and boundary-specific rather than binary.

Yet incomplete closure does not by itself determine which asset, if any, will occupy the residual space. Once closure mechanisms are recognized as boundary-selective rather than universally comprehensive, residual space cannot be treated as self-interpreting. The fact that no single mechanism fully internalizes every relevant boundary does not by itself indicate which instrument, liability form, or connective architecture will occupy the remaining space. At that point, the analytical problem shifts from closure capacity to selection capacity. Incomplete closure may preserve structural space, but structural space is not yet realized relevance. The next section therefore turns from closure mechanisms to the institutional constraints that may still block selection.

5. Layer-Lock and Selection Constraints

The existence of residual overflow does not by itself imply the selection of any particular external connective asset. Structural space is not the same as realized adoption. Even where competing closure mechanisms remain incomplete, external assets may still fail to be chosen because the relevant market participants operate under governance constraints, compliance requirements, liquidity preferences, distribution asymmetries, and operational incentives that favor other solutions. The analytical task, therefore, is not only to identify where overflow remains, but also to explain why selection may still fail.

One reason is governance lock. Institutions do not choose settlement instruments in a purely friction-minimizing vacuum. They choose within mandates, approval structures, legal interpretations, counterparty policies, and internal risk frameworks. An asset may appear economically useful at the system level while remaining difficult to approve at the institutional level. Governance can therefore function as a selection filter that blocks otherwise plausible connective instruments, not because the overflow problem has disappeared, but because admissibility is determined through internal decision structures.

A second constraint is compliance lock. Even if a connective asset could in principle reduce residual settlement friction, it may remain unusable or unattractive if its treatment across regulatory, custody, reporting, or counterparty frameworks is inconsistent. Institutions may prefer suboptimal but clearly recognized instruments over potentially more efficient but less uniformly admissible ones. Compliance asymmetry therefore does not merely reduce speed of adoption; it can redirect selection toward instruments that fit existing legal and supervisory categories, even when those instruments do not fully close the underlying overflow.

A third constraint is liquidity-depth lock. External connective assets compete not only on theoretical function but on execution reliability. If liquidity is too shallow, too fragmented, too volatile, or too stress-sensitive, institutions may refuse to depend on the asset in operationally meaningful form even when structural space exists for it. This matters because selection is ultimately conditioned by usable depth, not

conceptual elegance. A bridge-like role that fails under size, stress, or slippage constraints will not easily survive institutional scrutiny.

A fourth constraint is distribution lock. Some closure mechanisms benefit from immediate installed distribution through bank networks, issuer ecosystems, regulated customer bases, state channels, or existing treasury platforms. External connective assets often begin with weaker embedded distribution inside institutional workflows. This asymmetry matters because a technically plausible solution may still lose selection simply because alternative mechanisms arrive pre-integrated into the organizational and commercial environments where adoption decisions are made.

A fifth constraint is operational lock. Institutions tend to prefer workflows that are simple to integrate, observable to control functions, and legible to treasury, compliance, and audit teams. Even where an external asset could reduce a deeper structural friction, it may still fail to be selected if doing so introduces unfamiliar operational dependencies, new custody requirements, additional reconciliation steps, or uncertain failure modes. Selection is therefore shaped not only by what solves a problem in principle, but by what can be embedded with acceptable operational burden.

A sixth constraint is system-internal incentive lock. Competing settlement stacks often have reasons to preserve activity, balances, and control within their own perimeters. Banks may prefer bank-native liabilities. Stablecoin issuers may prefer issuer-centered circulation. Treasury systems may prefer internal optimization over external dependence. Market infrastructures may prefer solutions that deepen their own embedded role. This means that external connective assets may face resistance not because they are functionally irrelevant, but because existing systems possess strategic incentives to internalize flows within their own architectures wherever possible.

These constraints imply that the selection problem cannot be reduced to pure functional superiority. An asset may sit in a structurally plausible position while repeatedly losing actual selection to alternatives that are more governable, more admissible, more liquid at required scale, more deeply distributed, or easier to integrate. In that sense, residual overflow creates opportunity space, but opportunity space remains filtered through institutional selection mechanisms that can be highly conservative and path dependent.

This also helps explain why expansion, overflow, and selection must remain analytically distinct. Expansion may enlarge the addressable domain. Residual overflow may indicate that full closure has not been achieved. Yet selection may still fail because the institutional path from structural relevance to operational adoption is obstructed. The existence of that gap is not a minor implementation detail. It is one of the core reasons why external connective assets may remain perpetually plausible in theory while remaining weakly embedded in practice.

For this reason, the persistence of residual overflow should be interpreted as a necessary but insufficient condition for external connective-asset relevance. It identifies a field of possible necessity, not a completed outcome. Any stronger claim requires evidence that the asset not only occupies a plausible connective position, but overcomes governance, compliance, liquidity, distribution, operational, and incentive locks sufficiently to become selected in practice.

The implication for the present framework is straightforward. One cannot move directly from “closure is incomplete” to “the asset matters.” Between those claims lies a separate selection battle. To evaluate external connective assets seriously, that battle must be treated as a structural problem in its own right. The next step, therefore, is to specify the narrower conditions under which such assets may still matter, together with the conditions under which that relevance should be considered weakened or invalidated.

6. Conditions Under Which External Connective Assets Still Matter

External connective assets do not remain relevant simply because digital settlement infrastructures expand, nor simply because some residual friction persists. Relevance arises only under a narrower conjunction of conditions. The analytical question is not whether closure is incomplete in some abstract sense, but whether materially significant residual overflow survives after competing closure mechanisms operate, whether that overflow generates held-form demand, and whether any external asset is actually capable of surviving the institutional filters that govern selection.

Residual overflow is therefore necessary but not sufficient. Incomplete closure identifies a field of possible relevance, but not a completed outcome. Residual overflow matters only where the remaining frictions are economically material rather than merely formal, where they continue to affect settlement efficiency, collateral mobility, balance-sheet positioning, timing exposure, or operational resilience in a non-trivial way. If the remaining frictions are too small, too local, too compressible, or too easily internalized by bounded alternatives, then structural incompleteness does not by itself sustain a meaningful external connective role.

Held-form demand is also necessary but not sufficient. Even where residual overflow remains materially relevant, an external connective asset matters in the stronger sense only if those frictions generate reasons to retain the asset in operational form. Boundary-crossing settlement may require positioned balances and thus support inventory demand. Incomplete collateral portability may generate collateral demand. Staged or asynchronous workflows may generate waiting liquidity. Stress-sensitive execution environments may require liquidity-buffer demand. If no such retained-demand channel emerges, then visible connective function alone does not establish stronger relevance.

Selection survival is likewise necessary but not sufficient. Structural opportunity and retained-demand logic do not automatically translate into realized relevance if the asset cannot pass the governance, compliance, liquidity-depth, distribution, operational, and incentive constraints identified in Section 5. An external connective asset may be functionally plausible yet remain unselected in practice. For this reason, relevance depends not only on whether an unsolved boundary problem exists, but also on whether the asset can survive the actual institutional selection environment at meaningful scale.

A stronger relevance claim therefore arises only where three conditions converge in sequence rather than in isolation. First, residual overflow remains economically material after competing closure mechanisms operate. Second, that overflow generates held-form demand through inventory, collateral, liquidity-buffer, or waiting-liquidity channels. Third, the asset survives the institutional selection filters that govern realized adoption in actual workflows. These three conditions together identify a field of conditional relevance. None of them alone is enough.

These first three conditions are conjunctively necessary for any serious relevance claim: materially significant residual overflow, held-form demand, and survival through institutional selection filters. Comparative irreplaceability provides a further threshold. It is the test not merely of plausible connective relevance, but of stronger structural significance: whether removing the asset would materially worsen the relevant workflow relative to the actual alternatives available under those boundary conditions.

External connective assets do not need to dominate every workflow or outperform every alternative in every domain. But they matter in the stronger structural sense only where removing them would materially worsen cost, timing, slippage, collateral flexibility, liquidity efficiency, or operational resilience relative to the actual alternatives available in those boundary conditions. The relevant benchmark is therefore not abstract superiority, but comparative degradation under removal.

This derivation also clarifies the logic of weakening and invalidation. The relevance of an external connective asset weakens where residual overflow remains only formally present but economically immaterial, where held-form demand fails to emerge, or where institutional selection filters repeatedly block realization in practice. It is more strongly invalidated where competing closure mechanisms achieve sufficiently broad end-to-end coverage, where orchestration and liquidity-saving designs compress the relevant frictions to near-negligibility, or where no credible evidence appears that the asset is being retained in operationally meaningful form despite surrounding infrastructure growth.

Efficiency compression deserves particular weight here. Netting, synchronization, routing optimization, and liquidity-saving mechanisms may reduce the need for transitional inventory, collateral positioning, or waiting liquidity without eliminating all formal boundaries. In that case, the issue is not whether some friction remains in principle, but whether the remaining friction continues to impose material balance-sheet cost or operational vulnerability. Compression can therefore invalidate a stronger connective-asset thesis even without producing perfect closure. At the same time, compression should not be treated as costless abstraction: gains in capital efficiency may depend on coordination structures, legal linkages, or risk tolerances that are not universally available across institutions and jurisdictions. What matters is whether compression remains both effective and institutionally admissible.

The analytical implication is deliberately strict. Residual overflow is necessary but not sufficient. Held-form demand is necessary but not sufficient. Selection survival is necessary but not sufficient. Comparative irreplaceability is required for the stronger claim of structural significance. Only where these conditions converge, and only where substitution materially degrades relevant workflows, does an external connective asset remain structurally significant in the stronger sense. The framework is therefore deliberately restrictive: external connective-asset relevance is conditional rather than presumed, boundary-specific rather than general, and comparative rather than rhetorical.

7. Stress Extension to an XRP-Adjacent Context

This section is illustrative only. It does not function as evidentiary support for the general framework developed above, nor does it establish XRP-specific necessity. Its purpose is narrower: to show why an XRP-adjacent context remains a useful stress extension for the present argument, precisely because it concentrates many of the analytical risks that the framework is designed to discipline. Its role is not to import asset-specific conclusions into the framework, but to test whether the framework remains disciplined in a context where overreading pressures are unusually strong.

An XRP-adjacent context is useful because it sits close to a setting in which external connective-asset claims are especially vulnerable to overreading. Surrounding infrastructure may expand. Payment connectivity may improve. Tokenized cash instruments may grow. Treasury and custody layers may deepen. Yet none of these developments, by themselves, establishes that an external connective asset has become structurally relevant in the stronger sense defined in this paper. This makes the context analytically demanding rather than rhetorically convenient.

The first reason this context is useful as a stress extension is that it places expansion, closure competition, and asset-selection questions under unusually tight proximity. Stablecoins, tokenized deposits, CBDC-related pathways, internal treasury optimization, and orchestration-based routing all create plausible closure alternatives. If the framework is too weak, it will collapse here into either ecosystem overreading or residual-gap romanticism. The XRP-adjacent case therefore tests whether the argument can still separate expansion from closure, closure from selection, and selection from stronger structural relevance when the surrounding discourse strongly pressures those boundaries.

The second reason is that this context makes bypass risk unusually visible. Surrounding institutional infrastructure may succeed while the external connective asset itself remains weakly retained, weakly selected, or functionally bypassed. That possibility is not peripheral to the framework; it is one of its core disciplines. In an XRP-adjacent setting, the distinction between surrounding success and asset-specific relevance cannot be treated as a footnote. It is the main analytical burden.

The third reason is that this context makes the role of selection constraints especially clear. Even if residual overflow remains across settlement stacks, governance, compliance, liquidity-depth, distribution, operational, and incentive locks may still prevent realized selection of any external asset in operationally meaningful form. The case is therefore useful not because it straightforwardly confirms external connective-asset relevance, but because it forces the framework to confront the possibility that plausible connective space may coexist with weak institutional selection.

For these reasons, an XRP-adjacent context should be read here as a stress environment for the framework rather than as confirmatory evidence. The relevant question is not whether XRP, Ripple-linked infrastructure, or adjacent systems are expanding in visible terms. The relevant questions are stricter: whether economically material residual overflow survives competing closure mechanisms; whether that overflow generates held-form demand; whether any external asset survives institutional selection filters; and whether removal would materially degrade the relevant workflow relative to actual alternatives.

This is precisely why the present paper stops short of a stronger XRP-specific conclusion. The framework developed here is meant to discipline that later inquiry, not to pre-empt it. If a later analysis is to claim stronger relevance for any external connective asset in an XRP-adjacent environment, it must do so under the conditions specified above and not by importing conclusions from ecosystem growth, infrastructure adjacency, or rhetorical association alone.

In that sense, the XRP-adjacent stress extension serves a methodological function. It shows that the framework remains usable even where the temptation to overread is greatest. That is also why the next step lies not in advocacy, but in a narrower inquiry into density, sizing, and the empirical conditions under which retained-demand claims become materially defensible. For that reason, the XRP-adjacent extension

should be read as a stress test of analytical discipline rather than as an abbreviated case for asset-specific necessity.

8. Conclusion

This paper has argued that the expansion of digital financial infrastructure should not be conflated with settlement closure. Stablecoins, tokenized deposits, CBDCs, internal treasury rails, omnibus structures, and orchestration mechanisms may all expand the settlement field in meaningful ways. But expansion is not yet the stronger condition in which economically meaningful residual overflow has been compressed to immateriality. The central analytical task is therefore not to observe infrastructure growth in the abstract, but to ask whether competing closure mechanisms actually eliminate the boundary-crossing frictions that could sustain a role for an external connective asset.

From that starting point, the paper made five main moves. First, it separated expansion from closure and argued that closure is boundary-specific, uneven, and conditional rather than automatic or binary. Second, it classified the main structural sources of residual bridge-like overflow across jurisdictional, ledger, balance-sheet, collateral, timing, institutional, compliance, and stress-related dimensions. Third, it examined the major competing closure mechanisms not as abstract innovations, but as stack-specific attempts to internalize particular forms of friction, each with different strengths and different unresolved boundaries. Fourth, it argued that even where residual overflow persists, realized relevance may still fail because governance, compliance, liquidity-depth, distribution, operational, and incentive constraints function as independent selection barriers. Fifth, it specified the stricter conditions under which any external connective asset may still matter, together with the conditions under which that relevance weakens or is invalidated.

The resulting framework is deliberately restrictive. Residual overflow is necessary but not sufficient. Held-form demand is necessary but not sufficient. Selection survival is necessary but not sufficient. Comparative irreplaceability is required for the stronger claim of structural significance. External connective assets remain relevant in the stronger sense only where these conditions converge and where substitution would materially worsen the relevant workflow relative to available alternatives. Conversely, the case weakens where closure mechanisms achieve effective compression, where full-coverage substitutes absorb the relevant functions, where selection repeatedly fails, or where no credible evidence of retained demand emerges.

This paper does not claim that external connective assets necessarily survive institutional digitalization. Nor does it claim that expanding infrastructure automatically eliminates them. Its narrower contribution is to specify the conditions under which either conclusion becomes warranted. In that sense, the paper reframes the problem of external connective assets from one of adoption rhetoric to one of conditional structural relevance under closure competition.

The broader implication is methodological. If institutionally connected digital assets are to be evaluated seriously, they must be evaluated not by visible use, ecosystem growth, or interoperability claims alone, but by whether economically material residual overflow survives after competing closure mechanisms operate, whether that overflow generates held-form demand, and whether any asset can survive the institutional filters of actual selection. This does not settle the fate of any one asset. It does, however, establish a more disciplined framework for asking the question.

In the broader retained-demand audit series, this paper serves as the closure-competition layer: it tests whether the structural space identified in Paper 1 remains open after competing settlement stacks attempt to internalize residual overflow. That question must be answered before any later liquidity-density, sizing, or backend-retention analysis can be treated as meaningful.

The XRP-adjacent stress extension served here only to illustrate that the framework remains usable even where the temptation to overread is greatest. Any stronger asset-specific claim must come later and must be earned under the conditions specified in this paper rather than imported from ecosystem adjacency or infrastructure visibility. That later inquiry belongs not to the present paper, but to a further analysis of density, sizing, and the empirical conditions under which retained-demand claims become materially defensible.

Footnotes

1. This paper is designed to extend the analytic architecture of Paper 1 rather than to replace it. Paper 1 separated usage from retained demand and distinguished domain expansion, asset selection, and asset indispensability. The present paper asks under what structural conditions that retained-demand framework remains relevant once competing closure mechanisms expand across institutionally connected settlement environments.

2. Residual overflow, as used in this paper, does not refer to any remaining imperfection as such. It refers only to frictions that remain economically material after competing closure mechanisms operate. The term is introduced as an audit concept for institutionally connected digital finance in order to distinguish formally unresolved boundaries from economically consequential ones.

3. Comparative irreplaceability is used here as a stricter threshold than mere functional plausibility. An external connective asset need not dominate every workflow to matter in the stronger sense. What matters is whether removing it would materially worsen cost, timing, slippage, collateral flexibility, liquidity efficiency, or operational resilience relative to the actual alternatives available under the relevant boundary conditions.

Keywords

Retained Demand; Settlement Stacks; Closure Competition; Residual Overflow; External Connective Assets; Held-Form Demand; Asset Selection; Stablecoins; Tokenized Deposits; CBDC; Financial Market Infrastructure; Structural Audit; XRP

Author Note

This paper is presented as a working paper and should be read as a framework-oriented contribution. It is the second paper in a retained-demand audit series for institutionally connected digital assets. Its purpose is to clarify how external connective assets may be evaluated once infrastructure expansion, settlement closure, residual overflow, institutional selection, and stronger structural significance are treated as analytically distinct questions. XRP appears here only as a motivating case and illustrative stress

extension; the contribution of the paper is methodological rather than advocative. The paper does not provide investment advice, a price target, or a recommendation to buy, sell, or hold any digital asset.

Short Description

A working paper proposing a conditional structural framework for evaluating external connective assets under settlement stack competition. The paper distinguishes expansion from closure, residual overflow from asset selection, and held-form demand from stronger structural significance. It uses an XRP-adjacent context only as an illustrative stress extension, not as a price thesis.